CSS ANIMATIONS

1. What is CSS Animation?

CSS animation allows you to **change element styles smoothly over time** without using JavaScript.

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2. Key Concepts

a) `@keyframes`

Defines the animation steps.

```
@keyframes slide {
  from {
    transform: translateX(0);
  }
  to {
    transform: translateX(200px);
  }
}
```

---

b) Apply Animation to Element

```
.box {
  animation-name: slide;
  animation-duration: 2s;
}
```

---

3. Animation Properties (IMPORTANT)

1. animation-name

```
animation-name: slide;
```

---

2. animation-duration

Time animation takes to complete

```
animation-duration: 2s;
```

---

3. animation-timing-function

Controls speed curve

```
animation-timing-function: ease;
```

Types:

- `linear` → same speed
- `ease` → slow-fast-slow
- `ease-in`

- `ease-out`
- `ease-in-out`

---

4. animation-delay

Delay before animation starts

```
animation-delay: 1s;
```

---

5. animation-iteration-count

How many times animation runs

```
animation-iteration-count: infinite;
```

---

6. animation-direction

```
animation-direction: alternate;
```

Types:

- `normal`
- `reverse`
- `alternate`
- `alternate-reverse`

---

7. animation-fill-mode

Defines final state

```
animation-fill-mode: forwards;
```

Types:

- `none`
- `forwards`
- `backwards`
- `both`

---

8. animation-play-state

```
animation-play-state: paused;
```

---

4. Shortcut Property (Shorthand)

```
animation: slide 2s ease 1s infinite alternate;
```

Order:

```
name duration timing-function delay iteration-count direction
```

---

5. Percentage-Based Animation

```
@keyframes colorChange {
  0% {
    background-color: red;
  }
  50% {
    background-color: yellow;
  }
  100% {
    background-color: green;
  }
}
```

---

6. Example: Fade Animation

```
@keyframes fade {
  from {
    opacity: 0;
  }
  to {
    opacity: 1;
  }
}

.box {
  animation: fade 2s ease-in-out;
}
```

---

7. Example: Slide Animation

```
@keyframes slideIn {
  from {
    transform: translateX(-100%);
  }
  to {
    transform: translateX(0);
  }
}

.box {
  animation: slideIn 1s ease-out;
}
```

---

8. Example: Loading Spinner

```
.loader {
  width: 50px;
  height: 50px;
  border: 5px solid lightgray;
  border-top: 5px solid black;
  border-radius: 50%;
  animation: spin 1s linear infinite;
}

@keyframes spin {
  to {
    transform: rotate(360deg);
  }
}
```

---

9. Transform + Animation (IMPORTANT FOR INTERVIEW)

```
@keyframes move {
  0% {
    transform: translateX(0) scale(1);
  }
  100% {
    transform: translateX(200px) scale(1.2);
  }
}
```

---

10. Transition vs Animation

Feature	Animation	Transition
Control	Full control	Simple
Steps	Multiple (@keyframes)	Only start → end
Trigger	Auto	Needs event (hover, click)

---

11. Performance Tips (VERY IMPORTANT)

- Use `transform` instead of `top/left`
- Use `opacity` for smooth effects
- Avoid heavy animations on large elements
- Use `will-change` carefully

```
.box {
  will-change: transform;
}
```

---

12. Common Interview Questions

Q1: Difference between animation and transition?

→ Animation is automatic, transition needs trigger.

Q2: What is keyframes?

→ Defines intermediate steps of animation.

Q3: Which properties are best for performance?

→ `transform` and `opacity`

Q4: What is infinite animation?

```
animation-iteration-count: infinite;
```

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13. Real-World Use Cases

- Image sliders
- Loaders
- Hover effects
- Buttons animation
- Page transitions

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14. Pro Tip (IMPORTANT)

Use **combined animations**:

```
animation: fade 1s ease-in, slideIn 1s ease-out;
```

Image Slider with Fade Animation (Pure CSS)

Features

- Automatic sliding
- Smooth fade effect
- Responsive structure
- No JavaScript required

---

1. HTML

```
<div class="slider">
  <div class="slides">

    <div class="slide">
      <img src="https://picsum.photos/id/1015/800/400" />
    </div>

    <div class="slide">
      <img src="https://picsum.photos/id/1016/800/400" />
    </div>

    <div class="slide">
      <img src="https://picsum.photos/id/1018/800/400" />
    </div>

  </div>
</div>
```

---

2. CSS

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
```

```
.slider {
  width: 800px;
  height: 400px;
  margin: 40px auto;
  overflow: hidden;
  position: relative;
}

.slides {
  position: relative;
  width: 100%;
  height: 100%;
}

.slide {
  position: absolute;
  width: 100%;
  height: 100%;
  opacity: 0;
  animation: fade 9s infinite;
}

.slide img {
  width: 100%;
  height: 100%;
  object-fit: cover;
}

/* Delay for each slide */
.slide:nth-child(1) {
  animation-delay: 0s;
}

.slide:nth-child(2) {
  animation-delay: 3s;
}

.slide:nth-child(3) {
  animation-delay: 6s;
}

/* Fade animation */
@keyframes fade {
  0% { opacity: 0; }
  10% { opacity: 1; }
  30% { opacity: 1; }
  40% { opacity: 0; }
  100% { opacity: 0; }
}
```

How It Works

- All slides are positioned on top of each other using `position: absolute`
- Each slide has the same animation but with different delays
- The `@keyframes fade` controls when each slide appears and disappears
- Total animation duration is 9 seconds, divided equally among slides

---

Optional Enhancement: Pause on Hover

```
.slider:hover .slide {
  animation-play-state: paused;
```

```
}
```

Optional Enhancement: Responsive

```
.slider {
  width: 100%;
  max-width: 800px;
  height: auto;
  aspect-ratio: 2 / 1;
}
```

Interview Explanation (Short)

You can say:

“This slider uses CSS keyframes to animate opacity. All images are stacked, and each one is given a different animation delay to create a looping fade effect without JavaScript.”